

$\text{Def } \sum_{j=1}^{\infty} \tilde{a}_j$, полученный перестановкой
 менов $\sum_{k=1}^{\infty} a_k$, по перестановке пере $\sum_{k=1}^{\infty} a_k$

$(T_3) \quad \exists \sum_{k=1}^{\infty} a_k - \text{сх. о.в.с.} \Rightarrow \forall \text{ пер-ка}$
 $\sum_{j=1}^{\infty} \tilde{a}_j - \text{сх. о.в.с. к той же сумме.}$

$$\Delta \quad 1) \quad \exists \underline{a_k > 0} \Rightarrow \tilde{S}_n = \sum_{j=1}^n \tilde{a}_j$$

$$\forall j : 1 \leq j \leq n \quad \exists k_j : \tilde{a}_j = a_{k_j}$$

$$\exists l = \max(k_1, k_2, \dots, k_n) \rightarrow$$

$$\underline{\tilde{S}_n} = \sum_{j=1}^n \tilde{a}_j \leq \sum_{k=1}^l a_k \leq \underline{S}$$

$$\Rightarrow \{ \tilde{S}_n \} - \text{суп.} \Rightarrow \left\{ \begin{array}{l} \exists \lim S_n = \tilde{S} \\ \tilde{S} \leq S \end{array} \right. *$$

$$\text{Т.к. } \left\{ \begin{array}{l} \tilde{S} = \sum_{j=1}^{\infty} \tilde{a}_j \\ \sum_{k=1}^{\infty} a_k - \text{пер-ка } \sum_{j=1}^{\infty} \tilde{a}_j \end{array} \right. \xRightarrow{\uparrow} S \leq \tilde{S} **$$

$$\text{из } * \text{ и } ** \Rightarrow \tilde{S} = S$$

$$2) \quad \exists \underline{a_k \in \mathbb{R}}, \quad \exists a_k^+ = \begin{cases} a_k, & a_k > 0 \\ 0, & a_k \leq 0 \end{cases}$$

$$a_k^- = \begin{cases} -a_k, & a_k < 0 \\ 0, & a_k \geq 0 \end{cases}$$

$$\Rightarrow \left\{ \begin{array}{l} \sum a_k^+ \\ \sum a_k^- \end{array} \right. - \text{сх. о.в.с.} \Rightarrow \sum_{k=1}^{\infty} a_k = \sum_{k=1}^{\infty} a_k^+ - \sum_{k=1}^{\infty} a_k^-$$

$$= \sum_{j=1}^{\infty} (\tilde{a}_j)^+ - \sum_{j=1}^{\infty} (\tilde{a}_j)^- = \sum_{j=1}^{\infty} \tilde{a}_j \quad \blacktriangle$$

(T_4) Римана $\left[\sum_{k=1}^{\infty} a_k - \text{с.е. условно} \Rightarrow \forall l \in \overline{\mathbb{R}} \right]$
 $\exists \sum_{j=1}^{\infty} \tilde{a}_j = l \quad (\text{условно с.е.})$

Зам $\exists \sum_{j=1}^{\infty} \tilde{a}_j - \text{р.с.с.}, \text{ причём } \nexists \text{ сумма в } \overline{\mathbb{R}}.$

$\blacktriangle \exists \underline{l} \in \mathbb{R}$ Представим алгоритм перестановки!

$$\textcircled{D} \quad a_k^+ = \begin{cases} a_k, & a_k > 0 \\ 0, & a_k < 0 \end{cases} \quad a_k^- = \begin{cases} -a_k, & a_k < 0 \\ 0, & a_k \geq 0 \end{cases}$$

Т.к. $\{S_n^+\} \xrightarrow{n \rightarrow \infty} +\infty$, то $\exists n_1 \in \mathbb{N}$:

$$\left\{ \begin{array}{l} \sum_{k=0}^{n_1-1} a_k^+ \leq l \\ \sum_{k=0}^{n_1} a_k^+ > l \end{array} \right. \quad \left| \quad \begin{array}{l} a_0^+ = 0 \\ a_0^- = 0 \end{array} \right. \quad (1)$$

Т.к. $\{S_n^-\} \xrightarrow{n \rightarrow \infty} +\infty \Rightarrow \exists m_1 \in \mathbb{N}$:

$$\left\{ \begin{array}{l} \sum_{k=0}^{m_1-1} a_k^- \leq \sum_{k=0}^{n_1} a_k^+ - l \\ \sum_{k=0}^{m_1} a_k^- > \sum_{k=0}^{n_1} a_k^+ - l \end{array} \right. \quad (2)$$

$$\left\{ \begin{array}{l} \sum_{k=0}^{n_1} a_k^+ - \sum_{k=0}^{m_1-1} a_k^- > l \\ \sum_{k=0}^{n_1} a_k^+ - \sum_{k=0}^{m_1} a_k^- < l \end{array} \right. \quad (2')$$

Далее выберем $n_2 > n_1$:

$$\begin{cases} \sum_{k=0}^{n_1} a_k^+ - \sum_{k=0}^{m_1} a_k^- + \sum_{k=n_1+1}^{n_2-1} a_k^+ \leq L \\ \sum_{k=0}^{n_1} a_k^+ - \sum_{k=0}^{m_1} a_k^- + \sum_{k=n_1+1}^{n_2} a_k^+ > L \end{cases}$$

(if $n_2 = n_1 + 1$, то первое неравенство отсутствует)

Почему выбор возможен? - Т.к. $\sum_{k=n_1+1}^n a_k^+ \xrightarrow{n \rightarrow \infty} \infty$

Ан-то: т.к. $\sum_{k=m_1+1}^m a_k^- \xrightarrow{m \rightarrow \infty} +\infty$, то $\exists \underline{m_2 > m_1}$!

$$\begin{cases} \sum_{k=0}^{n_1} a_k^+ - \sum_{k=0}^{m_1} a_k^- + \sum_{k=n_1+1}^{n_2} a_k^+ - \sum_{k=m_1+1}^{m_2-1} a_k^- > L \\ \sum_{k=0}^{n_1} a_k^+ - \sum_{k=0}^{m_1} a_k^- + \sum_{k=n_1+1}^{n_2} a_k^+ - \sum_{k=m_1+1}^{m_2} a_k^- < L \end{cases}$$

(if $m_2 = m_1 + 1$, то 1^е нер-во отсут-т)

и т.д., т.е. пошер-но выбираем номера:

$$n_1 < n_2 < \dots < n_e < \dots \quad \text{и}$$

$$m_1 < m_2 < \dots < m_e < \dots$$

$$\text{Т.к. } \sum_{k=1}^{\infty} a_k^+ - \text{p.l.s.}, \quad \sum_{k=1}^{\infty} a_k^- - \text{p.l.s.}, \text{ то}$$

процессе выбора первое не может $\Rightarrow$

$$\exists \{n_e\}, \{m_e\} : \forall e \in \mathbb{N}$$

$$\forall n : \underline{n_e + 1 \leq n < n_{e+1}} \quad \text{вын:}$$

$$\begin{cases} \sum_{k=0}^{n_1} a_k^+ - \sum_{k=0}^{m_1} a_k^- + \dots - \sum_{k=m_{e-1}+1}^{m_e} a_k^- + \sum_{k=n_e+1}^n a_k^+ \leq L \\ \sum_{k=0}^{n_1} a_k^+ - \sum_{k=0}^{m_1} a_k^- + \dots - \sum_{k=m_{e-1}+1}^{m_e} a_k^- + \sum_{k=n_e+1}^{n_{e+1}} a_k^+ > L \end{cases}$$



1 if $n_{e+1} = n_e + 1$, то 1^е пер. отсчета - 1

$\forall e \in \mathbb{N} \quad \forall m: m_e + 1 \leq m < m_{e+1}$ верно:

$$\left\{ \begin{array}{l} \sum_{k=0}^{n_1} a_k^+ - \sum_{k=0}^{m_1} a_k^- + \dots + \sum_{k=n_e+1}^{n_{e+1}} a_k^+ - \sum_{k=m_e+1}^m a_k^- > 1 \\ \sum_{k=0}^{n_1} a_k^+ - \sum_{k=0}^{m_1} a_k^- + \dots + \sum_{k=n_e+1}^{n_{e+1}} a_k^+ - \sum_{k=m_e+1}^{m_{e+1}} a_k^- < 1 \end{array} \right. \quad \text{(*)}$$

1 if $m_{e+1} = m_e + 1$, то 1^е пер. отсчета - 1.

2 перестановку:

$$\sum_{j=1}^{\infty} \tilde{a}_j = \underbrace{a_1^+ + \dots + a_{n_1}^+}_{\text{}} - \underbrace{a_1^- + \dots + a_{m_1}^-}_{\text{}} +$$

$$+ \underbrace{a_{n_1+1}^+ + \dots + a_{n_2}^+}_{\text{}} - \underbrace{a_{m_1+1}^- + \dots + a_{m_2}^-}_{\text{}} + \dots$$

Вопрос: $\sum_{j=1}^{\infty} \tilde{a}_j = h$?

Разобьем $\mathbb{N}$ на отрезки:

$$[1, n_1 - 1], [n_1, n_1 + m_1 - 1], [m_1 + n_1, m_1 + n_2 - 1]$$

$$[n_2 + m_2, n_2 + m_2 - 1] \dots [m_e + n_e, m_e + n_{e+1} - 1]$$

$$[n_{e+1} + m_e, n_{e+1} + m_{e+1} - 1]$$

1 $\mathbb{N} \in [m_e + n_e, m_e + n_{e+1} - 1], e \in \mathbb{N}$

Следовательно $\mathbb{N} = m_e + n_e, n_e + 1 \leq n \leq n_{e+1} - 1 \Rightarrow$

2 $\sum_{k=1}^{\mathbb{N}} \tilde{a}_k = \sum_{k=1}^{n_1} a_k^+ - \sum_{k=1}^{m_1} a_k^- + \dots - \sum_{k=m_{e-1}+1}^{m_e} a_k^- + \sum_{k=n_e+1}^n a_k^+ \leq h$

$$\begin{aligned}
\Rightarrow \left| \sum_{k=1}^N \tilde{a}_k - h \right| &= h - \sum_{k=1}^N \tilde{a}_k = \\
&= h - \left[\sum_{k=1}^{n_1} a_k^+ - \sum_{k=1}^{m_1} a_k^- - \dots - \sum_{k=m_{e-1}+1}^{m_e-1} a_k^- - \boxed{a_{m_e}^-} \right. \\
&\quad \left. + \sum_{k=n_{e+1}}^{n_1} a_k^+ - \sum_{k=m_{e-1}+1}^{m_{e-1}} a_k^- \right] = h - \left[\sum_{k=1}^{n_1} a_k^+ - \sum_{k=1}^{m_1} a_k^- - \dots \right. \\
&\quad \left. - \sum_{k=m_{e-1}+1}^{m_{e-1}} a_k^- \right] - \sum_{k=n_{e+1}}^{n_1} a_k^+ + a_{m_e}^- \leq \\
&\quad \underbrace{\qquad\qquad\qquad}_{\geq h} \underbrace{\qquad\qquad\qquad}_{\geq 0} > 0
\end{aligned}$$

$$\leq h - h + a_{m_e}^- - 0 = a_{m_e}^-$$

(если $N = m_e + n_e$, $e \in \mathbb{N}$, то рассмотрим сумму в асимптотическом смысле $\Rightarrow$ устр. оценки верны)

$$\left| \sum_{k=1}^N \tilde{a}_k - L \right| \leq a_{m_e}^-, \quad N = m_e + n_e, \quad n_e \leq N \leq n_{e+1} - 1$$

(2) $] N \in [n_{e+1} + m_e, n_{e+1} + m_{e+1} - 1], e \in \mathbb{N}$

Если $N = n_{e+1} + m$, $m_{e+1} \leq m \leq m_{e+1} - 1 \Rightarrow$

$$\Rightarrow \sum_{k=1}^N \tilde{a}_k = \sum_{k=1}^{n_1} a_k^+ - \sum_{k=1}^{m_1} a_k^- + \dots + \sum_{k=n_{e+1}}^{n_{e+1}} a_k^+ - \sum_{k=m_{e+1}}^m a_k^- > h$$

$$\Rightarrow \left| \sum_{k=1}^N \tilde{a}_k - L \right| = \sum_{k=1}^N \tilde{a}_k - L =$$

$$= \left(\sum_{k=1}^{n_1} a_k^+ - \sum_{k=1}^{m_1} a_k^- + \dots + \sum_{k=n_{e+1}}^{n_{e+1}-1} a_k^+ \right) + \boxed{a_{n_{e+1}}^+} - L$$

$$- \sum_{k=m_{e+1}}^m \Big) - l \leq l + a_{n_{e+1}}^+ - 0 - l =$$

$$= a_{n_{e+1}}^+$$

(Если $N = n_{e+1} + m_e$, $e \in \mathbb{N}$, то пош. сумма в осязке $0 \leq T$, и осязке $0 \leq T$ верна)

$$\left| \sum_{k=1}^N \tilde{a}_k - L \right| \leq a_{n_{e+1}}^+$$

$$N = n_{e+1} + m, \quad m_e \leq m \leq m_{e+1} - 1$$

Т.к. $\begin{cases} \lim_{l \rightarrow \infty} n_{e+1} = \infty \\ \lim_{l \rightarrow \infty} m_e = \infty \end{cases} \Rightarrow \lim_{l \rightarrow \infty} a_{n_{e+1}}^+ = \lim_{l \rightarrow \infty} a_{m_e}^- = 0$

$$\Rightarrow \lim_{N \rightarrow \infty} \sum_{k=1}^N \tilde{a}_k = L \Rightarrow \sum_{k=1}^{\infty} \tilde{a}_k = L.$$



$$\lim_{n \rightarrow \infty} a_n = 0 \quad \forall \epsilon$$